Day 1

Thursday, 21 May 2026 8:42 AM

.c → Source code

↓

.out | .exe → compiled one

↓

0100111 00 → Dependent on OS, processor

.py

↓

.pyc

↓

PVM

↓

01010101

↓

Platform independent

Datatypes
- ↳ int
- ↳ float
- ↳ str
- ↳ bool.

Operators

Arithmetic → +, -, /, %, *, //, **

Relational → >= <= == !=

+= -= *= /= **=

%= //=

Logical → and, or, not.

n 60 → Divisible by 3 & 5

n 27 → Divisible by 3

n 35 → Divisible by 5

n 4 → Not divisible by 3 & 5

inclusive

for i in range (start, end) :   for i in range (start, end, step) :

exclusive

for i in range (end) :

n = 4

```
_ _ _ *
_ _ * *
_ * * *
* * * *
```

n = 3

```
_ _ *
_ * *
* * *
```

① No of rows
② No of cols in ith row
③ What needs to be printed?

n = 4

```
_ _ _ *  → 1
_ _ * *  → 2
_ * * *  → 3
* * * *  → 4
```

i	spaces	stars
1	3	1
2	2	2
3	1	3
4	0	4

① → n

② → n

③ → spaces + stars → n

spaces → $n - i$

stars → i

```python
n = int(input("Enter n : "))

for i in range(1, n + 1) :
    # spaces -> n - i
    for j in range(1, n - i + 1):
        print("_", end = " ")
    # stars -> i
    for j in range(1, i + 1) :
        print("*", end = " ")
    print()
```

n = 4

1 3 5 7
3 5 7 1
5 7 1 3
7 1 3 5

n = 5

1 3 5 7 9
3 5 7 9 1
5 7 9 1 3
7 9 1 3 5
9 1 3 5 7.

n = 4

1 3 5 7 → 1
3 5 7 1 → 2
5 7 1 3 → 3
7 1 3 5 → 4

$\boxed{7.}$
end

end → 2n − 1

i start →

1 1
2 3
3 5
4 7.

2i − 1

```python
n = int(input("Enter n : "))
end = (2 * n) - 1
for i in range(1, n + 1) :
    num = (2 * i) - 1
    for j in range(1, n + 1) :
        print(num , end = " ")
        num += 2
        if num > end :
            num = 1
    print()
```

n = 3

1 3 5

3 5 1

5 1 3

n = 4

1 _ _ _ *
2 _ _ * * *
3 _ * * * * *
n * * * * * * *
3 _ * * * * *
2 _ _ * * *
1 _ _ _ *

spaces -> n - 1   stars -> 2i - 1

n = 3

_ _ *
_ * * *
* * * * *
_ * * *
_ _ *

```python
n = int(input("Enter n : "))
# First half
for i in range(1, n + 1) :
    # spaces -> n - i
    for j in range(1, n - i + 1) :
        print("_", end = " ")
    # stars -> 2 * i - 1
    for j in range(1, (2 * i)) :
        print("*", end = " ")
```

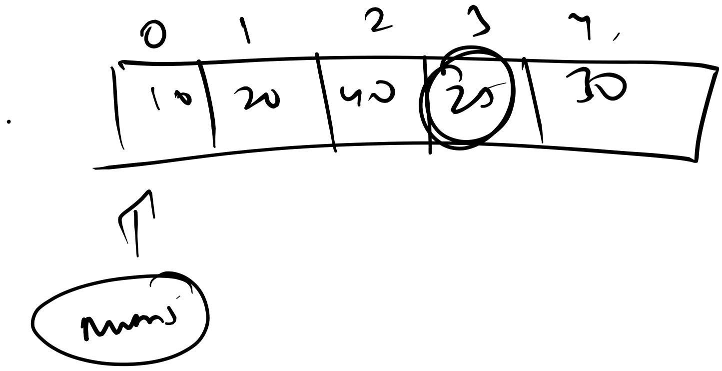
```python
# Second half
for i in range(n - 1, 0, -1) :
    # spaces -> n - i
    for j in range(1, n - i + 1) :
        print("_", end = " ")
    # stars -> 2 * i - 1
    for j in range(1, (2 * i)) :
        print("*", end = " ")
    print()
```

```
    print()
```

```
Enter n : 3
_ _ *
_ * * *
* * * * *
_ * * *
_ _ *
```

list → collection of elements

nums[3]




```
nums = [] → creating an empty list
n = int(input("Enter number of elements : "))
for i in range(n) :
    print("Enter ", i, " element : ", end = "")
    num = int(input())
    nums.append(num) → Adding num at last of list

# Accessing elements via forEach loop
for i in nums :
    print(i, end = " ")

# Accessing elements via index
# for i in range(len(nums)) :
#     print(i , " : ", nums[i])
```

```python
nums = [10,12,2,3]

nums.pop()  # Removes the last element, if list is empty throws IndexError

nums.remove(1) # Remove the first occurrence of the element, if element not
present, throws ValueError

nums1 = nums.copy() # Creates a new list and copy all elements from original
list
print(nums1)
nums.clear()    # Remove all the elements from list

nums.reverse() # Reverses the list

nums1 = [90,80,70]
nums.extend(nums1) # will add elements at last from object
```

```python
nums.insert(10, 90) # Inserts element at index, then all elements from index
-> rightShifted by 1 position, if index is not valid -> element will be
inserted at last index

nums.append(567)    # Appends the element to the last
print(nums)

print(nums.index(100)) # If present, Returns the first occurence of element
(int), else -> throw ValueError
print(nums.count(1))  # Returns the number of times, element is present in
list (int)
```

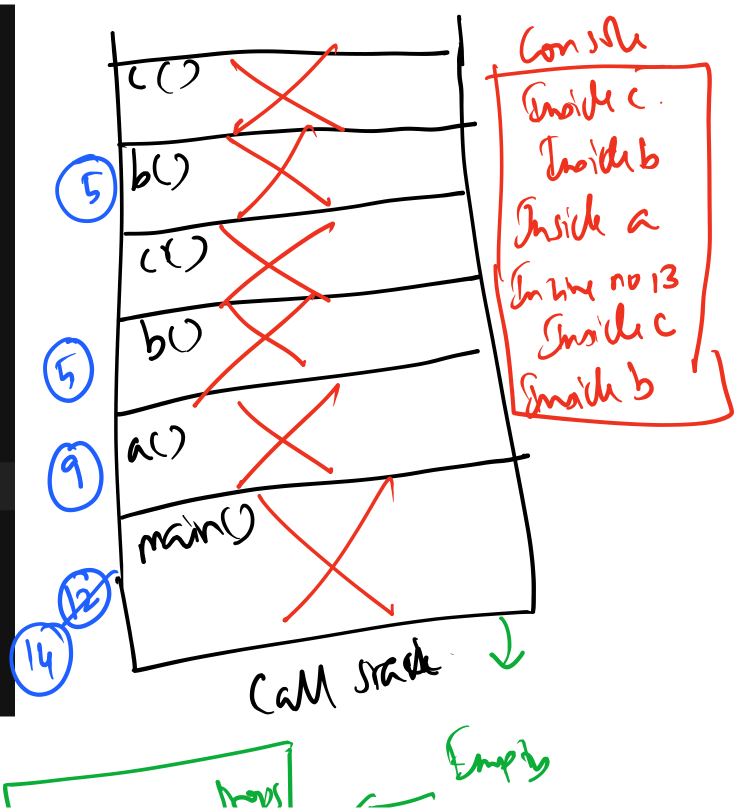
.

```python
def c() :
    print("Inside c")

def b() :
    c()
    print("Inside b")

def a() :
    b()
    print("Inside a")

a()
print("In line number 13")
b()
```



Program St

Program Steps

Call stack
↓
Empty

Call stack
↓
not Empty → 2 seconds
↓
TLE

0	1	2	3	4
~~10~~ 30	~~20~~ 25	40	~~25~~ 20	~~30~~ 10
i	i	i j	j	j

0 → 4   4 → 0

1 → 3   3 → 1

$\boxed{j = \text{len(nums)} - 1}$

```python
nums = [10,12,2,3,7,2,9]
# Reverse the list without inbuilt functions
i = 0
j = len(nums) - 1
while i < j :
    nums[i], nums[j] = nums[j], nums[i]
    i += 1
    j -= 1
print(nums)
```

→ swapping i & j elements.

TUPLE

```python
nums = (10,20,30)    # Tuple -> tuples are immutable,
modifying is not possible.
# Accessing all elements in tuple via forEach loop
for i in nums :
    print(i, end = " ")
# Iterative version of accessing elements in tuple
for i in range(len(nums)) :
    print(i , ":", nums[i])
nums[0] = 578  → Not possible in tuple, as elements are immutable.
print(nums.count(90))    # Count the number of times, element
is present -> Returns int
print(nums.index(100)) # Returns the first occurrence of
element, if not present throws ValueError -> Returns int
print(nums, type(nums))
```